Ruicheng Bao

Curriculum Vitae | Last updated: August 4, 2026

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Research Profile

Theoretical statistical physicist studying how randomness and intrinsic timescales shape nonequilibrium systems. My work spans stochastic thermodynamics, first-passage phenomena, relaxation and response, thermodynamic inference and optimal control, and open quantum dynamics. I am particularly interested in exact relations, universal scaling laws, and sharp bounds with clear operational meaning.

Academic Position

Apr. 2025–Present	**JSPS Research Fellow (DC1)**, Department of Physics, Graduate School of Science, The University of Tokyo Ito Laboratory, Tokyo, Japan

Education

Oct. 2024–Present	**Ph.D. program in Physics**, The University of Tokyo Advisor: Prof. Sosuke Ito
2024	**M.S. in Chemical Physics**, University of Science and Technology of China Advisor: Prof. Zhonghuai Hou
2021	**B.S. in Chemistry**, University of Science and Technology of China Advisor: Prof. Zhonghuai Hou

Publications

* Corresponding authors.

Unsupervised & Independent Work

- **Ruicheng Bao***, "Extreme First-Passage Time of Many Interacting Particles," arXiv:2607.22528 (2026).
- **Ruicheng Bao***, "Optimal Finite-Time Control of Nonreciprocal Brownian Dimers: Thermodynamic Anomaly and Multiple Transitions," arXiv:2607.20420 (2026).
- **Ruicheng Bao***, "Initial-State Typicality in Quantum Relaxation," Physical Review Letters 136, 070402 (2026).
- Caisheng Cheng and **Ruicheng Bao***, "Typical Mixing and Rare-State Bottlenecks in Open Quantum Systems," arXiv:2605.07619 (2026).
- Yijia Cheng, **Ruicheng Bao***, and Zhonghuai Hou*, "Hierarchical Reconstruction of Time-Arrow from Multi-Time Correlations," arXiv:2604.25749 (2026).
- Caisheng Cheng and **Ruicheng Bao**, "Physically Natural Metric-Measure Lindbladian Ensembles and Their Learning Hardness," arXiv:2601.01806 (2026).
- Xin Wang, **Ruicheng Bao**, and Naruo Ohga, "Characteristic Oscillations in Frequency-Resolved Heat Dissipation of Linear Time-Delayed Langevin Systems," Physical Review Research 8, 013039 (2026).
- **Ruicheng Bao*** and Shiling Liang*, "Nonlinear Response Identities and Bounds for Nonequilibrium Steady States," arXiv:2412.19602 (2024).

Supervised Work

- **Ruicheng Bao***, Naruo Ohga, and Sosuke Ito, "Measuring Irreversibility by Counting: A Random Coarse-Graining Framework," arXiv:2508.11586 (2025).
- **Ruicheng Bao*** and Zhonghuai Hou*, "Accelerating Quantum Relaxation via Temporary Reset: A Mpemba-Inspired Approach," Physical Review Letters 135, 150403 (2025).
- **Ruicheng Bao***, Chaoqun Du, Zhiyu Cao, and Zhonghuai Hou*, "Universal Trade-Off Between Irreversibility and Intrinsic Timescale in Thermal Relaxation with Applications to Thermodynamic Inference," Physical Review E 112, 044134 (2025).
- **Ruicheng Bao**, Zhiyu Cao, Jiming Zheng, and Zhonghuai Hou, "Designing Autonomous Maxwell's Demon via Stochastic Resetting," Physical Review Research 5, 043066 (2023); co-first authors.
- Zhiyu Cao, **Ruicheng Bao**, and Zhonghuai Hou, "Cascade-Enhanced Transport Efficiency of Biochemical Systems," Chaos 33, 063104 (2023); co-first authors.
- **Ruicheng Bao** and Zhonghuai Hou, "Improving Estimation of Entropy Production Rate for Run-and-Tumble Particle Systems by High-Order Thermodynamic Uncertainty Relation," Physical Review E 107, 024112 (2023).
- Zhiyu Cao, **Ruicheng Bao**, Jiming Zheng, and Zhonghuai Hou, "Fast Functionalization with High Performance in the Autonomous Information Engine," The Journal of Physical Chemistry Letters 14, 66–72 (2023); co-first authors.

Presentations

Talks

- "Initial-State Typicality in Quantum Relaxation," lightning talk, Workshop on Stochastic Thermodynamics VII (WOST VII), May 18–22, 2026 (Online).
- "Initial-State Typicality and Applications to Quantum Thermal Tasks," invited seminar, Hatano Laboratory, The University of Tokyo, Dec. 20, 2025, Tokyo, Japan.
- "Initial-State Typicality and Applications to Quantum Thermal Tasks," invited seminar, Sasa Laboratory, Kyoto University, Dec. 1, 2025, Kyoto, Japan.
- "An Exact Theory of Nonequilibrium Response to Arbitrarily Strong Perturbations," invited seminar, Sasa Laboratory, Kyoto University, Jun. 2, 2025, Kyoto, Japan.
- "Dissipation, Intrinsic Timescale and Coarse-Graining in Relaxation Processes," lightning talk, Workshop on Stochastic Thermodynamics V (WOST V), 2024 (Online).
- "Universal Trade-Off Between Irreversibility and Intrinsic Timescale in Thermal Relaxation," contributed talk, 28th International Conference on Statistical Physics (STATPHYS28), Aug. 7–11, 2023, The University of Tokyo, Tokyo, Japan.

Posters

- "Extreme First-Passage Time in Interacting Many-Body Systems: A No-Go Theorem and Beyond," Frontiers in Nonequilibrium Physics 2026, YITP, Kyoto University, May 13, 2026.
- "A Random Coarse-Graining Framework with Applications to Thermodynamic Inference," Kyoto Workshop on Quantum Thermodynamics and Stochastic Thermodynamics 2025, YITP, Kyoto University, Dec. 8, 2025.
- "Estimating Irreversibility under Coarse-Graining," Frontiers in Nonequilibrium Physics 2024, YITP, Kyoto University, Jul. 17, 2024.
- "Universal Trade-Off Relation Between Irreversibility and Intrinsic Timescale in Thermal Relaxation," Frontiers in Nonequilibrium Physics 2024, YITP, Kyoto University, Jul. 10, 2024.

Academic Activities

- Teaching assistance in statistical mechanics and physical chemistry.
- Referee for *Physical Review Letters*, *Physical Review A*, *Physical Review E*, *Journal of Mathematical Physics*, and *Physica A*.

Research Methods

Stochastic processes and first-passage theory; inequalities and asymptotic analysis; information geometry and information theory; stochastic analysis; open-system dynamics; analytical and computational modeling.